



Modelling Financial Markets

Seminar 10: Market Microstructure — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ ETF: Exchange-Traded Fund
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Execute orders on a limit order book and compute their cost step by step
- ▣ Derive spreads from the Roll, Glosten–Milgrom and Kyle models
- ▣ Compute an optimal execution schedule (Almgren–Chriss)
- ▣ Estimate spreads and illiquidity from real data, with bootstrap and HAC inference

- ▣ Background: Lecture 10; textbooks [Hasbrouck \(2007\)](#), [Bouchaud et al. \(2018\)](#), [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** nine pen-and-paper exercises (A1–A9)
- ▣ **Part B:** seven computational exercises with inference (B1–B7)
- ▣ **Part C:** an open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- ▣ Units: relative spreads and price moves in basis points ($1 \text{ bp} = 0.01\%$); traded value in USD
- ▣ Intraday data have strong dependence within a day: resample whole days, not single bars

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (a market buy walks the book), A3 (the Roll model), A5 (Glosten–Milgrom quotes), A7 (the Kyle equilibrium), A8 (Almgren–Chriss schedule)
- Part B: B1 (the intraday U-shape), B2 (spread estimators at two frequencies), B4 (illiquidity and the VIX)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (model: A5), A9 (model: A8)
- Part B: B3 (spreads across 25 assets; model: B2), B5 (Amihud on three markets; model: B2, B4), B6 (volume and price moves; model: B4), B7 (the Bitcoin clock; model: B1)
- Part C: C1, open question and project seed; C2, critique of an AI answer

Setup: Python Environment and Data

-  **Open in Colab:** `chapter10_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- SPY 5-minute bars, regular session, October 2020 – September 2026; Bitcoin 5-minute bars, September 2024 – September 2026
- Daily open, high, low, close and volume for 7 US stocks, 11 BVB stocks and 7 crypto-assets; VIX; BET-TR (BET with dividends reinvested)
- BVB traded values converted from lei to USD at the BNR reference rate of each day

Formulas Used Today

- Mid-price $m = (a + b)/2$; relative spread $(a - b)/m$; cost of a buy of average price $\bar{p}$: $(\bar{p} - m)/m$
- Roll Roll (1984): $\text{Var}(\Delta p) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$, $s = 2c$
- Glosten–Milgrom Glosten & Milgrom (1985): $a = V_L + (V_H - V_L)P(V_H \mid \text{buy})$, $b = V_L + (V_H - V_L)P(V_H \mid \text{sell})$, Bayes' rule
- Kyle Kyle (1985): $\beta = \sigma_u/\sigma_v$, $\lambda = \sigma_v/(2\sigma_u)$, expected profit $\sigma_v\sigma_u/2$
- Almgren–Chriss Almgren & Chriss (2001): $x_k = X \sinh(\kappa(T - t_k))/\sinh(\kappa T)$, $\cosh(\kappa\tau) = 1 + \tilde{\kappa}^2\tau^2/2$, $\tilde{\kappa}^2 = \lambda\sigma^2/\tilde{\eta}$, $\tilde{\eta} = \eta - \gamma\tau/2$
- Amihud Amihud (2002): $\text{ILLIQ} = \frac{1}{D} \sum_d |r_d|/\text{DVOL}_d$

A1: A Market Buy Walks the Book [Solved]

Task

Bids		Asks	
400	99.98	100.02	300
600	99.97	100.03	500
800	99.95	100.05	400
1 500	99.90	100.08	1 000

- (a) A market order buys 1 000 shares: which prices and quantities execute?
- (b) Average price, cost relative to the mid-price (in USD and bp) and to the best ask
- (c) The book after the trade: new best ask and spread

Solution

- 300 at 100.02, 500 at 100.03, 200 at 100.05
- $\bar{p} = (300 \cdot 100.02 + 500 \cdot 100.03 + 200 \cdot 100.05) / 1\,000 = 100.031$; total 100 031 USD
- Mid $m = 100.00$: cost 0.031 USD per share = 3.1 bp; above the best ask: 0.011 USD
- New best ask 100.05 with 200 shares left; spread $100.05 - 99.98 = 0.07$ (7 bp, from 4 bp)

A2: A Market Sell, Then a Limit Buy [Proposed]

Solution discussed in class

Task

Bids		Asks	
400	99.98	100.02	300
600	99.97	100.03	500
800	99.95	100.05	400
1 500	99.90	100.08	1 000

- (a) A market order sells 1 500 shares on the same book: fills, average price, cost relative to the mid in bp
- (b) Then a limit buy of 500 shares at 99.99 arrives: new best bid, spread and mid-price
- Model: A1



A3: The Roll Model, Step by Step [Solved]

Task

- A stock trades around 20 lei; daily price changes have $\text{Var}(\Delta p) = 0.0520$ and $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -0.0081$ (lei²) Roll (1984)
- (a) Half-spread c , spread s in lei and in % of the price
- (b) Variance of the efficient price change σ^2 ; share of $\text{Var}(\Delta p)$ due to the bounce
- (c) First-order autocorrelation of Δp

Solution

- $c = \sqrt{0.0081} = 0.090$ lei, $s = 2c = 0.18$ lei, i.e. 0.9% of 20 lei
- $\sigma^2 = 0.0520 - 2c^2 = 0.0520 - 0.0162 = 0.0358$
- Bounce share $2c^2/\text{Var}(\Delta p) = 31.2\%$
- $\rho_1 = -0.0081/0.0520 = -0.156$
- Almost a third of the daily variance is noise: volatility from trade prices overstates the efficient volatility

A4: Roll with Another Stock [Proposed]

Solution discussed in class

Task

- ▣ A stock trades around 12 lei; $\text{Var}(\Delta p) = 0.0300$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -0.0030$
- ▣ (a) c , s in lei and in %; σ^2 ; bounce share; ρ_1
- ▣ (b) In another year the sample covariance is $+0.0020$: what does the Roll estimator give, and why can this happen?
- ▣ Model: A3



A5: Glosten–Milgrom Quotes [Solved]

Task

- $V_L = 90$, $V_H = 110$, $\theta = P(V_H) = 0.5$; share of informed traders $\mu = 0.3$; uninformed traders buy or sell with probability $1/2$ Glosten & Milgrom (1985)
- (a) $P(\text{buy} \mid V_H)$ and $P(\text{buy} \mid V_L)$
- (b) Ask, bid and spread
- (c) The first trade is a buy: new θ and new quotes

Solution

- $P(\text{buy} \mid V_H) = 0.3 + 0.35 = 0.65$, $P(\text{buy} \mid V_L) = 0.35$
- $P(V_H \mid \text{buy}) = \frac{0.65 \cdot 0.5}{0.65 \cdot 0.5 + 0.35 \cdot 0.5} = 0.6500 \Rightarrow$
 $a = 90 + 20 \cdot 0.6500 = 103.00$
- $P(V_H \mid \text{sell}) = 0.3500 \Rightarrow b = 97.00$; spread
 $6.00 = \mu(V_H - V_L)$
- After a buy: $\theta = 0.6500$; $P(V_H \mid \text{buy}) = 0.7752$:
 $a = 105.50$, $b = 100.00$, spread 5.50
- Quotes move up after a buy and the spread narrows: the market maker has learned something



A6: Glosten–Milgrom with More Informed Traders [Proposed]

Solution discussed in class

Task

- ▣ $V_L = 90$, $V_H = 110$, prior $\theta = 0.7$, $\mu = 0.5$
- ▣ (a) Ask, bid and spread; is the spread still $\mu(V_H - V_L)$?
- ▣ (b) Two buys in a row: θ and quotes after each
- ▣ Model: A5

A7: The Kyle Equilibrium [Solved]

Task

- Value $v \sim N(p_0, \sigma_v^2)$ with $\sigma_v = 2$ USD; noise order flow $\sigma_u = 10\,000$ shares [Kyle \(1985\)](#)
- (a) Insider intensity β and price impact λ
- (b) Expected insider profit; how much of σ_v^2 remains after trading?
- (c) Total order flow $y = +15\,000$ shares: price change

Solution

- $\beta = \sigma_u / \sigma_v = 5\,000$ shares per USD of mispricing
- $\lambda = \sigma_v / (2\sigma_u) = 2 / 20\,000 = 0.0001$ USD per share (1 cent per 100 shares)
- Expected profit $\sigma_v \sigma_u / 2 = 10\,000$ USD
- $\text{Var}(v \mid p) = \sigma_v^2 / 2 = 2$: half of the information stays private
- $\Delta p = \lambda y = 0.0001 \cdot 15\,000 = 1.50$ USD

A8: Almgren–Chriss Schedule, Step by Step [Solved]

Task

- Sell $X = 10^6$ shares in $T = 5$ days, $N = 5$ ($\tau = 1$); $\sigma = 0.95$, $\eta = 2.5 \cdot 10^{-6}$, $\gamma = 2.5 \cdot 10^{-7}$, $\lambda = 2 \cdot 10^{-6}$ Almgren & Chriss (2001)
- (a) $\tilde{\eta}$, $\tilde{\kappa}^2$ and κ
- (b) Holdings x_k and daily sales n_k
- (c) Expected cost and standard deviation vs selling evenly

Solution

- $\tilde{\eta} = 2.5 \cdot 10^{-6} - 1.25 \cdot 10^{-7} = 2.375 \cdot 10^{-6}$;
 $\tilde{\kappa}^2 = \frac{2 \cdot 10^{-6} \cdot 0.9025}{2.375 \cdot 10^{-6}} = 0.76$
- $\cosh \kappa = 1 + 0.76/2 = 1.38 \Rightarrow \kappa = 0.8463$, $\sinh(5\kappa) = 34.40$
- $x = (1\,000\,000; 428\,599; 182\,933; 76\,296; 27\,643; 0)$
- $n = (571\,401; 245\,666; 106\,637; 48\,652; 27\,643)$: 57% on day 1
- $\mathbb{E}[C] = 1.078$, $\text{sd}[C] = 0.449$ million USD; even selling: 0.600 and 1.041

A9: A More Risk-Averse Seller [Proposed]

Solution discussed in class

Task

- Same order as A8, but $\lambda = 2 \cdot 10^{-5}$
- (a) $\tilde{\kappa}^2$, κ and the holdings x_k
- (b) Expected cost and standard deviation; compare with A8
- (c) When would a trader choose this schedule?
- Model: A8

B1: The Intraday U-Shape of SPY [Solved]

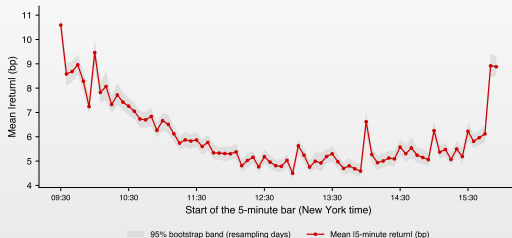
Task

- SPY 5-minute bars, 1 477 full days
- (a) Mean $|r|$ in the first bar (09:30) and in the midday bars (11:30–14:00); their ratio
- (b) Share of daily volume traded in the last 30 minutes
- (c) 95% bootstrap intervals, resampling whole days (1 000 resamples) [Efron \(1979\)](#)

Interpretation

- Why resample days instead of single bars?
- Is the U-shape a feature of every day or an average of a few extreme days?
- What does it imply for a trader who must buy 1 million shares today?

B1: Results [Solved]



- Mean $|r|$: 10.6 bp in the first bar, 5.1 bp at midday: ratio 2.06, 95% CI [1.98, 2.15]
- Last 30 minutes: 18.6% of daily volume, 95% CI [18.3, 19.0]%, against $6/78 = 7.7\%$ if volume were even
- Bars within a day share the day's volatility: resampling days keeps that dependence
- The narrow intervals show a systematic pattern; a large buyer should avoid the open and use the close (or the closing auction)

B2: Spread Estimators at Two Frequencies [Solved]

Task

- SPY: Roll, Corwin–Schultz and Abdi–Ranaldo Roll (1984), Corwin & Schultz (2012), Abdi & Ranaldo (2017)
- (a) From 5-minute bars, per day, averaged over days; 95% CI resampling days
- (b) From daily bars, September 2024 – September 2026; 95% CI by a moving-block bootstrap (blocks of 20 days)
- (c) Compare with one price step (USD 0.01) relative to the price

Interpretation

- Which frequency gives a credible spread?
- What do the daily estimators measure instead?
- Why is the Abdi–Ranaldo interval so wide?

B2: Results [Solved]

	Roll	Corwin–Schultz	Abdi–Ranaldo
5-minute bars (bp)	2.82 [2.09, 3.43]	3.11 [3.03, 3.20]	1.06 [0.00, 1.67]
Daily bars (bp)	64.3	22.4 [17.4, 29.4]	55.1 [0.0, 92.6]

- One price step: 0.21 bp; even the 5-minute Corwin–Schultz estimate is 15 times larger
- Daily estimators are two orders of magnitude too high: with a daily standard deviation of 103.4 bp, the high-low range is almost all volatility
- Abdi–Ranaldo is the square root of a mean that is often negative (set to 0): its interval touches 0
- Roll: the 5-minute covariance is negative on only 59% of days

B3: Spread Estimates Across 25 Assets [Proposed]

Solution discussed in class

Task

- ▣ 7 US stocks, 11 BVB stocks, 7 crypto-assets; daily bars, September 2024 – September 2026
- ▣ (a) Corwin–Schultz and Abdi–Rinaldo for each asset; group medians
- ▣ (b) Spearman rank correlation of Corwin–Schultz with annualised volatility; of the Amihud ratio with median traded value
- ▣ Model: B2

Interpretation

- ▣ Do BVB stocks look more or less liquid than US stocks?
- ▣ Which measure would you trust to compare markets, and why?
- ▣ What would you need to validate either measure?

B4: Illiquidity and the VIX [Solved]

Task

- Daily intraday illiquidity of SPY: mean of $|r|/\text{DVOL}$ over the 78 bars, bp per USD 100 million; 1 477 days
- (a) OLS regression $\ln \text{ILLIQ}_d = a + b \ln \text{VIX}_d + e_d$
- (b) Standard errors: OLS and HAC with 20 lags
Newey & West (1987)
- (c) Add a linear time trend (years): does b change?

Interpretation

- Interpret b as an elasticity
- Why do the two standard errors differ so much?
- Is the relation causal?

B4: Results [Solved]

- ▣ $\hat{b} = 0.95$; OLS SE 0.038, HAC SE 0.121; 95% CI (HAC) [0.72, 1.19]; $R^2 = 0.30$
- ▣ A 10% higher VIX goes with about $0.95 \times 10\%$ higher illiquidity: nearly proportional
- ▣ Residuals have first-order autocorrelation 0.76: the HAC standard error is 3.2 times the OLS one
- ▣ With a trend: $\hat{b} = 0.80$ (SE 0.120); trend -0.062 per year (SE 0.021): illiquidity falls over time, the VIX link remains
- ▣ Not causal: both respond to news and uncertainty; market makers widen quotes when risk rises (Section 5 of the lecture)

B5: Amihud Illiquidity on Three Markets [Proposed]

Solution discussed in class

Task

- ▣ Same 25 assets; Amihud ratio (bp per USD 1 million) over September 2024 – September 2026 [Amihud \(2002\)](#)
- ▣ (a) Group medians; ratio BVB / US with a 95% bootstrap CI (resample assets within each group)
- ▣ (b) Rank stability: Spearman correlation between the rankings of the first and the second year
- ▣ Model: B2, B4

Interpretation

- ▣ How much less liquid is the BVB, and how precise is that number?
- ▣ Is illiquidity a stable characteristic of an asset?
- ▣ What drives the width of the interval?

B6: Volume and Price Moves in SPY Bars [Proposed]

Solution discussed in class

Task

- SPY 5-minute bars; relative volume = bar volume / median volume at the same time of day; $z = |r| /$ daily s.d. of 5-minute returns
- (a) 20 groups by relative volume; slope of $\ln \bar{z}$ on $\ln$ volume; 95% CI resampling days
- (b) Bar-level regression of $\ln |r|$ on $\ln$ relative volume with standard errors clustered by day; test slope = 0.5
- Model: B4

Interpretation

- Is the relation concave? Is it a square-root law [Tóth et al. \(2011\)](#)?
- Why can bar data not test the metaorder law?
- Why cluster by day?

B7: The Bitcoin Clock [Proposed]

Solution discussed in class

Task

- ▣ Bitcoin 5-minute bars, 749 days (UTC)
- ▣ (a) Weekdays: mean $|r|$ in 13:00–17:00 UTC (US morning) vs 03:00–11:00 UTC; ratio with a 95% CI resampling days
- ▣ (b) Weekday vs weekend mean $|r|$; ratio with a 95% CI
- ▣ Model: B1

Interpretation

- ▣ Why does a 24/7 market have a clock?
- ▣ When would you execute a large Bitcoin order, and why?

C1: Does Illiquidity Carry a Premium on the BVB?

Question

- How illiquid is the BVB compared with US and crypto markets, and is illiquidity priced?
- Amihud (2002): expected illiquidity should **raise** future returns; an unexpected rise should **lower** current returns [Amihud \(2002\)](#)
- Data: monthly BET-TR, SPY and Bitcoin returns; market illiquidity from the stocks (tokens) of each market, 2016–2026

What a good answer contains

- A market illiquidity index (mean log Amihud) and its AR(1) shock
- Regression of monthly returns on lagged illiquidity and the shock, with HAC standard errors
- A cross-section on the BVB: illiquid minus liquid stocks, rebalanced every year
- Few stocks, ten years: how much power does any test have? [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have daily closes P (lei) and volumes V (number of shares, positive every day) for a BVB stock. Estimate the Roll spread and the Amihud illiquidity. What does a round trip cost me?"

The AI answer: text

- Roll: $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$ and $s = 2c$ Roll (1984)
- If the covariance comes out positive, take its absolute value: the formula still gives the spread
- Amihud: the average of $|r_d|$ divided by the volume of day d , per 1 million Amihud (2002)
- Result: $s = 0.05$ lei on a 20-lei stock, i.e. 0.25% of the price; a round trip (buy, then sell) costs $2s$, i.e. 0.50%

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np
dp = P.diff().dropna().values
g = np.cov(dp[1:], dp[:-1])[0, 1]
s_roll = 2 * np.sqrt(abs(g))
s_pct = 100 * s_roll / P.mean()
r = np.log(P).diff().dropna()
illiq = 1e6 * (r.abs() / V.loc[r.index]).mean()
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ What sign of the autocovariance does the Roll model predict? What does the opposite sign tell you?
- ▣ Compare with the definition of $ILLIQ$ on the Formulas frame: what is $DVOL_d$?
- ▣ Draw the bid, the ask and the mid-price: at which prices do you buy and sell?

Common Pitfalls

- ▣ Reading a daily high-low spread estimate as the true spread of a liquid asset
- ▣ Comparing illiquidity across markets without converting the traded value to one currency
- ▣ Treating 5-minute bars of the same day as independent observations
- ▣ Using raw prices and volumes across a stock split: the traded value jumps
- ▣ Calling a volume–volatility relation in bars a price-impact law

Take-Home Assignment (Team Project)

- Choose three assets not covered in class (a BVB stock outside the ten blue chips, a crypto token, a US small cap)
- Compute Amihud and the two high-low spread estimators by month; plot them with the volatility
- Estimate the cost of selling a position of USD 1 million with the square-root law, using each asset's volume and volatility
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Applied Computational Finance

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

References

- ▣ Abdi, F., Rinaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies*, 30(12), 4437–4480.
- ▣ Almgren, R., Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.
- ▣ Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
- ▣ Bouchaud, J.-P., Bonart, J., Donier, J., Gould, M. (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.
- ▣ Brauneis, A., Mestel, R., Riordan, R., Theissen, E. (2021). How to measure the liquidity of cryptocurrency markets? *Journal of Banking & Finance*, 124, 106041.
- ▣ Corwin, S.A., Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance*, 67(2), 719–760.
- ▣ Efron, B. (1979). Bootstrap methods: another look at the jackknife. *Annals of Statistics*, 7(1), 1–26.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Glosten, L.R., Milgrom, P.R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100.
- ▣ Hasbrouck, J. (2007). *Empirical Market Microstructure*. Oxford University Press.
- ▣ Kyle, A.S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Pele, D.T., Mazurencu-Marinescu, M., Nijkamp, P. (2013). Herding behaviour, bubbles and log periodic power laws in illiquid stock markets: a case study on the Bucharest Stock Exchange. SSRN Working Paper 2305945.
- ▣ Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4), 1127–1139.
- ▣ Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2), 021006.